

Alexander DeLise

Tampa, FL | 813-291-6177 | ✉ alex.r.delise@gmail.com | 💻 [alexanderdelise](https://alexanderdelise.com) | 🎓 [Google Scholar](https://scholar.google.com/citations?user=9333333333333333) | 🔗 [alexdelise](https://github.com/alexdelise)

EDUCATION

Florida State University

May 2027

B.S. Applied and Computational Mathematics, B.S. Computational Science, GPA - 4.0

Tallahassee, FL

- **Awards:** Presidential Scholar, Florida Bright Futures Scholar, Pi Mu Epsilon Mathematics Honor Society
- **Relevant Coursework:** Computational Probabilistic Modeling, Data Mining and Machine Learning, High Performance Computing, MCMC Methods, Discrete/Continuous Algorithms for Science, Math Modeling, Statistics
- **Honors in the Major Thesis:** Generative AI and Compressed Sensing, advised by Dr. Nick Dexter

INDUSTRY EXPERIENCE

Johns Hopkins University Applied Physics Laboratory

November 2025 – Present

Incoming Data Science Intern

Laurel, MD

- KBS Systems Performance Analysis Group

RESEARCH EXPERIENCE

Florida State University

November 2025 – Present

Undergraduate Researcher – Generative AI and Compressed Sensing

Tallahassee, FL

- Researching novel applications of generative and diffusion models with applications in compressed sensing
- Developing theoretical guarantees for generative compressed sensing and validating them through large-scale experiments on medical imaging and natural-image datasets

Emory University

June 2025 – August 2025

Research Fellow – Scientific Machine Learning and Data Science | 🔗

Atlanta, GA

- Developed Bayesian methods for optimal rank-constrained encoder-decoder neural networks, connecting to dimensionality reduction and inverse problems
- Implemented and trained autoencoders in PyTorch on biomedical imaging and financial datasets, benchmarking against theoretically optimal mappings
- Performed large-scale data preprocessing, feature extraction, and model evaluation (MSE, RMSE, explained variance), identifying significant reconstruction gaps in standard training approaches, and even deep neural networks
- Collaborated in a diverse team, honing communication, adaptability, and joint problem-solving to drive research progress
- First author on manuscript on Bayesian machine learning and neural networks, under submission to a peer-reviewed journal

University of Tennessee, Knoxville

May 2024 – April 2025

Research Fellow – Quantum Algorithms | 🔗

Knoxville, TN

- Built preprocessing pipeline to identify and prepare feasible solutions for constrained binary optimization problems
- Developed a quantum-state-labeling method with ancillary variables, achieving up to 360× improvement in solution quality compared to baseline heuristics
- Implemented and benchmarked quantum optimization algorithms with custom state preparation using Python/PennyLane
- Designed a reproducible evaluation suite by serializing parameter configurations into callable files for scalable experiments
- Coordinated within a cross-disciplinary group to troubleshoot, share insights, and reach common goals for steady progress
- Co-authored a manuscript demonstrating method under submission to a peer-reviewed journal

FAMU-FSU College of Engineering

September 2023 – June 2025

Undergraduate Researcher – Operations Research | 🔗

Tallahassee, FL

- Designed and implemented large-scale optimization models for pandemic-era patient allocation and healthcare facility planning across Florida's 67 counties
- Processed and analyzed extensive COVID-19 datasets (hospitalization, vaccination, geospatial) and integrated them with epidemic SIRV dynamics and policy constraints into a decision-making framework
- Simulated 1 million+ patient transfers, achieving a 16.4% reduction in unmet hospital demand, identifying bottlenecks
- Developed pipelines in Python, R, CPLEX, and GAMS for data processing, model implementation, and data visualization
- First author on a paper accepted to the 2025 North American IEOM Conference in Orlando, FL

TECHNICAL SKILLS

Languages: Python, R, MATLAB, C++, SQL, LaTeX

Tools: Git, GitHub, VS Code, IBM ILOG CPLEX, GAMS, Gurobi, Microsoft Office Suite

Frameworks and Libraries: NumPy, PyTorch, pandas, SciPy, scikit-learn, gurobipy, TensorFlow, PennyLane, tidyverse, ggplot

Involvements: *Participant*, JPMorganChase Data for Good Hackathon | Society for Industrial and Applied Mathematics